AMIDI: Audio-Supervised Event Modeling for Expressive MIDI Generation

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1 The expressiveness gap of symbolic models

Symbolic models offer editable music, but typical training targets contain less performance information than audio.

Symbolic music

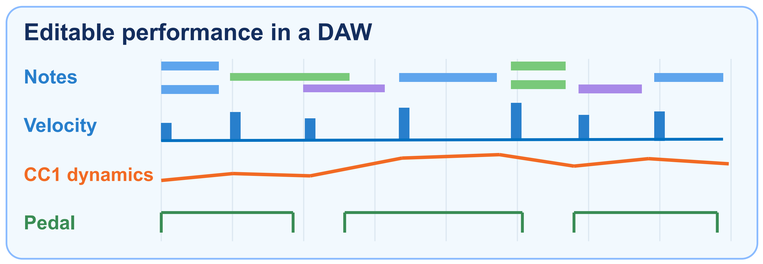
Cons: Limited performance cues
Pros: Note-level editing

Audio

Cons: Difficult to edit individual notes
Pros: Rich performance detail

2 What professional musicians need

Our goal: a symbolic model that generates notes, velocity, CC1 dynamics curves, and sustain-pedal events.



Musicians can refine the generated performance while arranging, using the same controls they already edit.

3 Scarcity in expressive training data

Expressive MIDI–audio pairs are scarce at scale.
Lakh MIDI provides scale, but controller coverage is uneven. MAESTRO provides detailed performance MIDI, but mainly covers classical piano.

4 AMT alone is insufficient

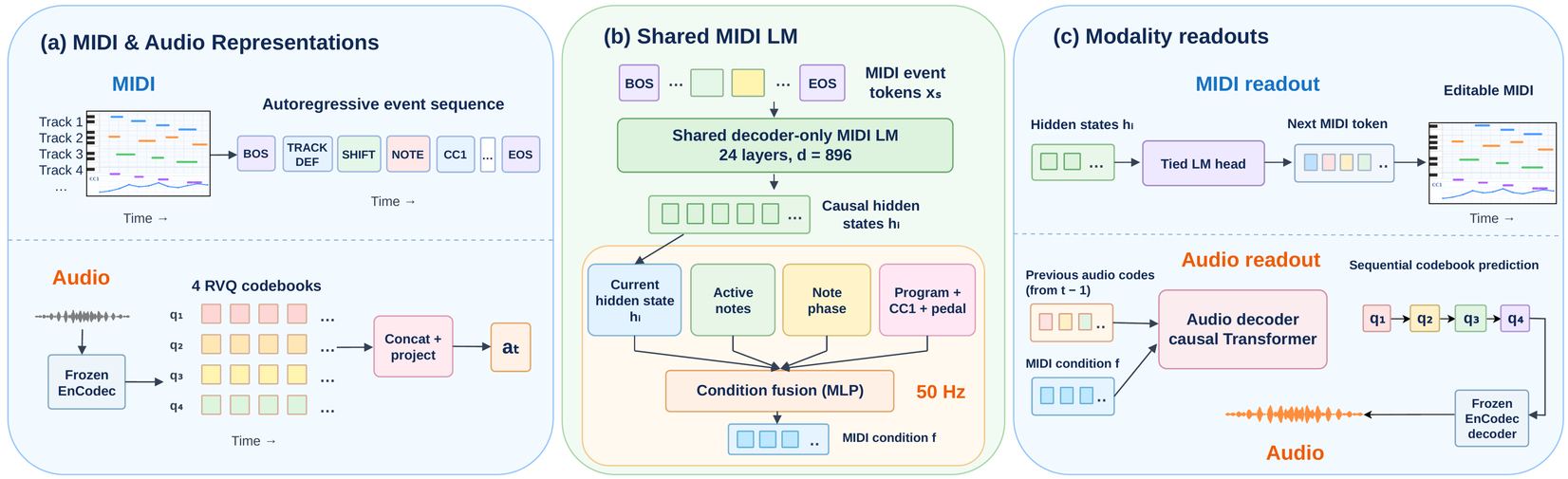
GiantMIDI-Piano uses automatic music transcription (AMT) to build MIDI datasets from recordings.

Audio → AMT → Pseudo-MIDI

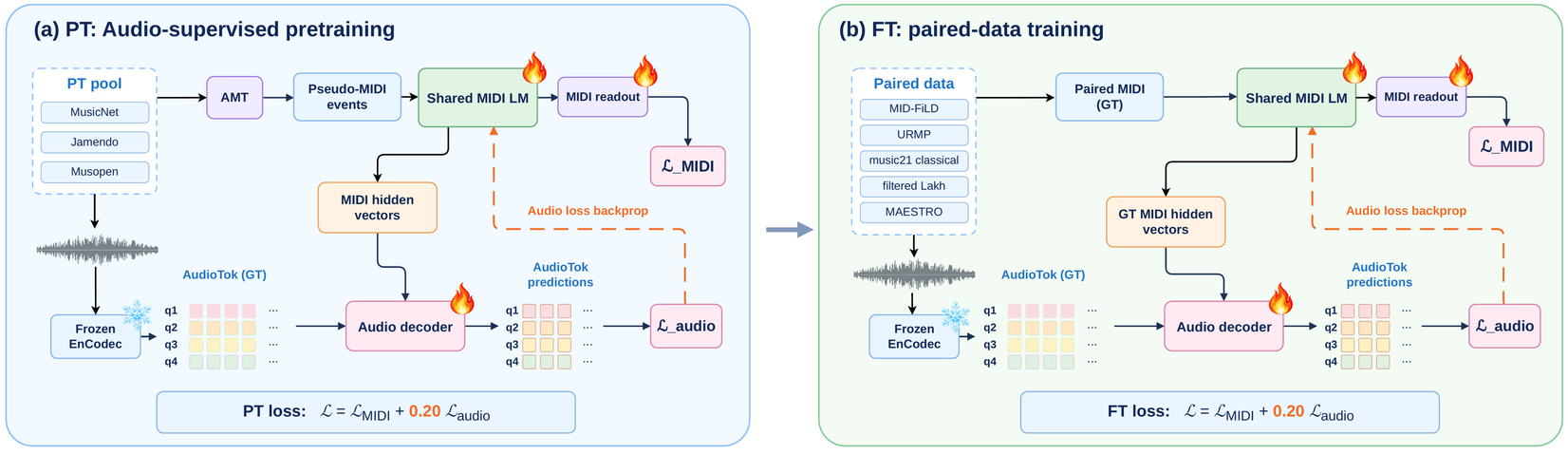
Transcription errors remain. AMT does not reliably recover continuous dynamics and performance details, so pseudo-MIDI can sound mechanical.

Our approach: use the original audio as an additional supervision signal during pretraining.

5 AMIDI Model Architecture



6 Experimental training pipeline



Use audio during pretraining, then fine-tune on a smaller set of MIDI–audio pairs. Only the experimental group receives audio supervision during PT; both groups use the same paired fine-tuning.

8 Demo

7 Pretraining results

Comparable best-checkpoint fit suggests no loss of MIDI predictive performance, with a small potential benefit from audio supervision.

